

SPEAKER NOTES

Gene Therapy & Genetic Engineering

MBBS / College-Level Presentation · 15 Slides

SLIDE 1 TITLE SLIDE

Gene Therapy and Genetic Engineering

SPEAKER NOTES

Welcome the audience. Introduce yourself and the topic. Mention this presentation covers gene therapy and genetic engineering at a college/MBBS level, touching on mechanisms, applications, ethics, and future scope.

DELIVERY TIPS

- Set a confident, enthusiastic tone from the outset.
- Briefly state your name, institution, and the purpose of this talk.
- Optional: ask the audience a quick show-of-hands question — 'Who has heard of CRISPR?'

SLIDE 2 INTRODUCTION

Definitions of Gene Therapy & Genetic Engineering

SPEAKER NOTES

Define both terms clearly. Emphasize: genetic engineering is the technique; gene therapy is the clinical application. Give analogy: genetic engineering is like software coding; gene therapy is running the patch on a sick computer (body).

DELIVERY TIPS

- Spend time on the key distinction — students often confuse the two terms.
- Use the software analogy: GE = programming language, Gene Therapy = the actual patch applied to fix a bug.
- Mention that this presentation will cover both dimensions — tools AND clinical applications.

KEY POINTS FOR STUDENTS

1. Gene Therapy: introduction of therapeutic DNA/RNA into cells to treat or prevent disease.
2. Genetic Engineering: direct manipulation of an organism's genome using biotechnology tools.
3. Key link: GE provides the toolbox; Gene Therapy is its targeted medical application.

SLIDE 3 HISTORY & DEVELOPMENT

Key Milestones from 1953 to 2023

SPEAKER NOTES

Walk through the timeline. Highlight 1953 (foundation), 1990 (first trial), 2012 (CRISPR revolution), and 2023 (real-world CRISPR approval). Emphasize how rapidly the field has advanced.

DELIVERY TIPS

- Pause at each landmark — don't rush through 70 years of science.

- 1953: Watson & Crick — foundational discovery, Nobel Prize 1962.
- 1990: Ashanthi DeSilva (ADA-SCID) — first approved human gene therapy trial.
- 2012: Doudna & Charpentier publish CRISPR-Cas9 — Nobel Prize 2020.
- 2023: FDA approves Casgevy — first CRISPR therapy for sickle cell disease.

🔍 KEY POINTS FOR STUDENTS

4. 70 years of progress from double helix to approved CRISPR therapy.
5. Each decade brought a transformative breakthrough.
6. The pace of innovation has accelerated dramatically since 2010.

SLIDE 4 TYPES OF GENE THERAPY

Somatic vs. Germline Gene Therapy

🔍 SPEAKER NOTES

Distinguish somatic vs germline clearly. Somatic is ethical and approved. Germline is controversial and banned in most countries. Mention He Jiankui controversy (2018 CRISPR baby scandal). Ask students: why is germline therapy so debated?

🔍 DELIVERY TIPS

- Draw a clear comparison table on the whiteboard if needed.
- He Jiankui case (2018): Chinese researcher edited CCR5 gene in embryos to confer HIV resistance — jailed 3 years.
- Ask the class: 'If germline editing could eliminate cancer from your bloodline forever, would you allow it?'
- Somatic therapy is body-cell targeted — changes do NOT pass to children.
- Germline therapy alters egg/sperm/embryo — changes pass to ALL future generations.

🔍 KEY POINTS FOR STUDENTS

7. Somatic: clinically approved, ethically accepted — targets body cells only.
8. Germline: banned in most countries — changes inherited by future generations.
9. He Jiankui scandal (2018) remains the most controversial germline experiment in history.

SLIDE 5 METHODS OF GENE DELIVERY

Viral Vectors & Non-Viral Methods

🔍 SPEAKER NOTES

Explain viral vs non-viral delivery. AAV is the gold standard for clinical gene therapy. Lipid nanoparticles gained fame via COVID mRNA vaccines. Discuss the trade-off: viral = efficient but immune response risk; non-viral = safer but less efficient.

🔍 DELIVERY TIPS

- AAV (Adeno-Associated Virus): most widely used, low immunogenicity, used in Zolgensma and Luxturna.
- Retroviruses/Lentiviruses: integrate permanently into host DNA — powerful but carry insertional mutagenesis risk.
- LNPs (Lipid Nanoparticles): became globally known via Pfizer-BioNTech and Moderna COVID vaccines.

- Key trade-off: viral = highly efficient delivery but immune reaction risk; non-viral = safer but lower efficiency.
- Mention electroporation as a commonly used ex vivo method in research labs.

🔍 KEY POINTS FOR STUDENTS

10. ~70% of current gene therapy trials use viral vectors.
11. AAV is the gold-standard vector — low immune response, tissue specific.
12. LNPs represent the breakthrough non-viral platform, enabling mRNA therapies.

SLIDE 6 GENETIC ENGINEERING TECHNIQUES

Recombinant DNA, CRISPR, Gene Cloning, TALEN/ZFN

🔍 SPEAKER NOTES

Explain recombinant DNA (the original genetic engineering tool), then CRISPR (the modern revolution). Cloning is foundational to all molecular biology. TALEN/ZFN are predecessors of CRISPR. Emphasize CRISPR is like a 'find and replace' function in DNA.

🔍 DELIVERY TIPS

- Recombinant DNA: restriction enzymes cut → gene inserted into plasmid → bacteria express protein. Insulin since 1982 made this way.
- CRISPR analogy: sgRNA = GPS/address, Cas9 = scissors — 'find and replace' in the genome.
- Gene cloning: foundational for PCR, diagnostics, and protein production at scale.
- TALEN & ZFN are older (2000s–2010s) — more expensive, complex, and now largely superseded by CRISPR.
- CRISPR won the Nobel Prize in Chemistry 2020.

🔍 KEY POINTS FOR STUDENTS

13. Recombinant DNA technology is the original GMO technique — responsible for insulin, vaccines, and diagnostics.
14. CRISPR-Cas9 is the most precise, affordable, and programmable gene-editing tool ever developed.
15. Gene cloning amplifies specific DNA sequences for research and therapeutic use.

SLIDE 7 CRISPR-CAS9 EXPLAINED

Mechanism: Step-by-Step with Diagram

🔍 SPEAKER NOTES

Walk through each step of CRISPR carefully. Use the analogy: sgRNA = GPS coordinates, Cas9 = molecular scissors. NHEJ is error-prone (useful for knockouts), HDR is precise (useful for gene correction). Mention real-world applications: sickle cell disease, cancer, blindness.

🔍 DELIVERY TIPS

- Step 1 — Design sgRNA: a 20-nucleotide 'address code' matching the target DNA.
- Step 2 — sgRNA + Cas9 form a complex: the active 'search and cut' machine.
- Step 3 — DNA Scanning: complex patrols the genome until it finds the matching sequence + PAM site.

- Step 4 — Double-Strand Break: Cas9 cuts BOTH strands of the DNA helix precisely.
- Step 5 — Repair pathways: NHEJ (fast, error-prone → gene knockout) vs. HDR (slow, precise → gene correction with template).
- Clinical examples: Casgevy (sickle cell), CAR-T (cancer), Luxturna (blindness).

🔍 KEY POINTS FOR STUDENTS

16. CRISPR uses a guide RNA to direct Cas9 protein to a precise genomic location.
17. NHEJ repair: introduces mutations — used to knock out disease-causing genes.
18. HDR repair: uses a template to insert correct gene sequence — used for gene correction therapies.

SLIDE 8 MEDICAL APPLICATIONS

Cancer, Inherited Diseases, Eye Disorders, Vaccines

🔍 SPEAKER NOTES

Cover the six major areas. Emphasize approved therapies: Casgevy, Zolgensma, Luxturna, CAR-T. Mention cost issue (Zolgensma = \$2.1M one-time dose). CAR-T cell therapy is the most exciting oncology breakthrough. mRNA vaccines brought gene delivery to mainstream awareness.

🔍 DELIVERY TIPS

- Blood disorders: Casgevy (CRISPR, FDA 2023) and Zolgensma (AAV9, \$2.1M) are landmark approvals.
- CAR-T cell therapy: patient's T-cells are genetically engineered to express chimeric antigen receptors targeting cancer cells.
- Luxturna: subretinal injection of AAV2 carrying RPE65 gene — restores vision in congenital blindness.
- mRNA vaccines: COVID-19 vaccines were the world's introduction to gene delivery medicine.
- HIV: researchers are targeting CCR5 receptor (the HIV entry point) with CRISPR to make cells HIV-resistant.
- Cost challenge: \$2.1M for Zolgensma — discuss who bears this cost and healthcare equity implications.

🔍 KEY POINTS FOR STUDENTS

19. Approved therapies: Casgevy, Zolgensma, Luxturna, Hemgenix, Kymriah, Yescarta.
20. CAR-T therapy represents the first successful cancer gene therapy platform.
21. mRNA vaccines validated lipid nanoparticle delivery on a global scale.

SLIDE 9 AGRICULTURAL APPLICATIONS

GM Crops, Pest Resistance, Nutritional Enhancement

🔍 SPEAKER NOTES

GM crops are grown on ~190 million hectares. Major products: Bt corn/cotton, Roundup Ready soy, Golden Rice. Discuss controversy: golden rice could prevent 500,000+ childhood blindness cases/year but faced regulatory delays. CRISPR in agriculture is not classified as 'GMO' in some countries, easing approval.

🔍 DELIVERY TIPS

- Bt crops: *Bacillus thuringiensis* toxin gene inserted — kills specific insects without pesticide sprays.
- Roundup Ready soy: glyphosate-resistant — simplifies weed control but raises herbicide dependency concerns.
- Golden Rice: contains beta-carotene (Vitamin A precursor) — could prevent blindness in millions of malnourished children.
- The golden rice controversy: 20+ years of regulatory delay despite humanitarian potential.
- CRISPR in agriculture is excluded from GMO regulations in the US, Japan, and Argentina — faster approval.
- Discuss environmental concerns: gene flow to wild plants, monoculture risks, biodiversity impact.

🔍 KEY POINTS FOR STUDENTS

22. ~190 million hectares of GM crops grown globally (2023).
23. Bt crops and Roundup Ready varieties are the most commercially successful GM products.
24. CRISPR-edited crops may bypass traditional GMO regulations in many countries.

SLIDE 10 ADVANTAGES & BENEFITS

Disease Prevention, Personalized Medicine, Economic Impact

🔍 SPEAKER NOTES

Highlight the transformative potential. Key message: gene therapy addresses the ROOT CAUSE. Personalized medicine is the future — matching therapy to individual genome. Emphasize economic benefits too: cheaper insulin production via recombinant DNA saved millions of lives.

🔍 DELIVERY TIPS

- Root cause treatment: unlike traditional drugs (that manage symptoms), gene therapy fixes the underlying defect.
- One-time treatment potential: Zolgensma — a single \$2.1M injection that lasts a lifetime vs. \$100K/year enzyme replacement.
- Personalized medicine: pharmacogenomics uses your genome to select the right drug at the right dose.
- Recombinant insulin: before 1982, diabetics used animal-derived insulin — GE transformed availability and purity.
- Vaccine speed: COVID mRNA vaccines developed in 9 months — traditional vaccines take 10–15 years.
- Research tools: CRISPR allows generation of disease models in weeks — accelerating drug discovery enormously.

🔍 KEY POINTS FOR STUDENTS

25. Gene therapy treats ROOT CAUSE — not just symptoms.
26. One-time curative potential changes the economics of medicine.
27. Personalized genomic medicine is the future of therapeutics.

SLIDE 11 RISKS & ETHICAL CONCERNS

Safety, Ethics, Regulatory Frameworks

🔍 SPEAKER NOTES

Discuss the Jesse Gelsinger case (1999): first gene therapy death, set back the field by years. He Jiankui case (2018): first CRISPR babies, led to international outrage and his imprisonment. Key ethical principles: non-maleficence, justice, autonomy. Ask: should germline editing ever be allowed?

🔗 DELIVERY TIPS

- Jesse Gelsinger (1999): 18-year-old died of immune reaction to adenoviral vector in OTC deficiency trial — FDA halted gene therapy trials for years.
- He Jiankui (2018): edited CCR5 in embryos → twin girls born with modified genome → jailed 3 years in China.
- Off-target effects: CRISPR can cut unintended sites — ongoing research to improve specificity.
- Insertional mutagenesis: early retroviral trials caused leukemia in ADA-SCID patients.
- Ethics framework: Autonomy (consent), Beneficence (do good), Non-maleficence (do no harm), Justice (fair access).
- Designer babies: slope toward selecting intelligence, appearance, athletic ability — eugenics concerns.
- Ask: 'If CRISPR could eliminate a genetic disease from your family forever, would you — and should you — use it?'

🔗 KEY POINTS FOR STUDENTS

28. Jesse Gelsinger death (1999) set the field back a decade — immune reaction to adenoviral vector.
29. He Jiankui scandal (2018) — first CRISPR babies — resulted in 3-year imprisonment and global condemnation.
30. Regulatory frameworks (FDA, EMA, WHO) exist but germline editing remains banned globally.

SLIDE 12 CURRENT RESEARCH & FUTURE SCOPE

AI in Genomics, Base Editing, In Vivo CRISPR, WGS

🔗 SPEAKER NOTES

AlphaFold is a landmark — it solved a 50-year grand challenge in biology. AI + CRISPR is the future. Prime editing is safer than standard CRISPR (no double-strand break). By 2030, expect gene therapies to become as routine as chemotherapy. WGS cost curve mirrors Moore's Law.

🔗 DELIVERY TIPS

- AlphaFold2 (DeepMind, 2021): predicted structures of ~200 million proteins — transforming drug discovery.
- Base editing (David Liu, Broad Institute): changes individual DNA bases (A→G, C→T) WITHOUT cutting — safer, more precise.
- Prime editing: 'search and replace' for any mutation — no double-strand break, no donor DNA needed.
- In vivo CRISPR: Intellia Therapeutics' NTLA-2001 delivered via LNPs directly into the liver — first in vivo CRISPR trial.
- WGS cost: from \$3 billion (2003) → \$1,000 (2014) → <\$200 (2024) — approaching routine clinical use.
- 2030 prediction: >50 approved gene therapies, CRISPR cures for HIV and common cancers, personalized cancer vaccines.

🔗 KEY POINTS FOR STUDENTS

31. AI (AlphaFold, machine learning) is transforming genomics and drug target identification.
32. Base editing and prime editing are safer next-generation alternatives to standard CRISPR.
33. By 2030, gene therapies are expected to be as routine as chemotherapy.

SLIDE 13 CASE STUDIES

ADA-SCID, Sickle Cell, Blindness, SMA — Real Success Stories

🔗 SPEAKER NOTES

These are the landmark cases every medical student should know. Ashanthi = proof of concept (1990). Luxturna = first approved (2017). Casgevy = CRISPR milestone (2023). Zolgensma = most expensive medicine, but cost-effective as a lifetime cure. Discuss: how do we decide who gets \$2M therapies?

🔗 DELIVERY TIPS

- Case 1 — Ashanthi DeSilva (1990): 4-year-old with ADA-SCID, no immune system. Retroviral vector delivered ADA gene into T-cells. She is alive and healthy 30+ years later.
- Case 2 — Casgevy (2023): Victoria Gray was the first sickle cell patient treated. CRISPR reactivated fetal hemoglobin. 94% of patients pain-crisis-free at 1 year.
- Case 3 — Luxturna (2017): AAV2 carrying RPE65 gene injected subretinally. First FDA-approved in vivo gene therapy. Patients can navigate in dim light independently.
- Case 4 — Zolgensma (2019): Single IV infusion of AAV9-SMN1. Most expensive drug ever (\$2.1M). 91% of treated infants alive at age 2, vs <8% of untreated infants.
- Discussion point: With a one-time cost of \$2.1M, how should healthcare systems decide who gets treated?
- Insurance perspective: \$2.1M once vs. \$100K/year lifetime enzyme replacement = cost-effective over 20+ years.

🔗 KEY POINTS FOR STUDENTS

34. Ashanthi DeSilva (1990) — first gene therapy success, still alive 30+ years later.
35. Casgevy (2023) — first FDA-approved CRISPR therapy for sickle cell disease.
36. Zolgensma (\$2.1M) — most expensive drug ever, yet arguably cost-effective over a lifetime.

SLIDE 14 CONCLUSION

Key Takeaways & Closing Remarks

🔗 SPEAKER NOTES

Summarize all six key points. Reinforce that gene therapy is not science fiction — it's here today. End with the quote and open for Q&A. Suggested question to pose: 'If you could eliminate a genetic disease from your family line, would you — and should you be allowed to?'

🔗 DELIVERY TIPS

- Recite the six takeaways clearly and deliberately — don't rush.
- Emphasize: gene therapy is NOT future medicine — it is present-day, approved medicine.
- Close with the discussion question to generate engagement: 'If you could eliminate a genetic disease from your family bloodline forever, would you — and should you be allowed to?'
- Thank the audience warmly and invite questions.
- Optional follow-up resources: genome.gov, FDA gene therapy page, WHO ethics framework.

🔗 KEY POINTS FOR STUDENTS

37. Gene therapy treats ROOT CAUSE — not symptoms.
38. CRISPR has revolutionized the field — precise, cheap, programmable.
39. Approved therapies prove the technology works in real patients.
40. Agriculture benefits from GMOs — food security, nutrition, sustainability.
41. Ethical governance is essential — access, equity, germline limits.
42. AI + genomics = the next frontier of personalised medicine.

SLIDE 15 REFERENCES

Primary Sources for Further Reading

SPEAKER NOTES

Direct students to these primary sources for further reading. Emphasize peer-reviewed journals (Science, Nature) and official health agencies (WHO, NIH, FDA) as gold-standard sources. Robbins Pathology has an excellent chapter on genetic disorders.

DELIVERY TIPS

- Emphasize: always cross-reference with peer-reviewed primary literature, not Wikipedia or blogs.
- Top recommended reads: Doudna & Charpentier 2012 (Science); Robbins Pathology Ch. 5; National Academies 2017 report.
- Online resources: genome.gov (NHGRI), fda.gov/gene-therapy, who.int/ethics.
- PubMed is the go-to database for gene therapy research — free, comprehensive.
- For students: The Code Breaker by Walter Isaacson (popular science book on CRISPR) is highly recommended.

KEY POINTS FOR STUDENTS

43. Primary sources: FDA, WHO, NIH, Nature, Science.
44. Robbins Pathology — best clinical reference for genetic disorders.
45. The Code Breaker (Isaacson) — excellent accessible read on CRISPR's history.